

Modbus

Industrial Control Systems · SCADA Hub

COURSE DESCRIPTION

Modbus is the most widely deployed fieldbus protocol in industrial automation, present in virtually every manufacturing and utility installation worldwide. This course covers RTU mode on RS-485, ASCII mode, and Modbus TCP over Ethernet, along with the register data model, function codes, exception handling, and data type encoding across vendor implementations. Security limitations and network-layer compensating controls are addressed in the context of modern OT network architecture.

PREREQUISITES

Basic TCP/IP networking concepts. No prior SCADA or industrial protocol experience required.

COURSE TOPICS

1. **Introduction:** Protocol origins, the 1979 register model, and why the original design persists unchanged in modern field devices.
2. **Data Model:** Coils, discrete inputs, input registers, and holding registers — table structure, data types, and access rules.
3. **RTU Mode:** RS-485 framing, CRC-16 integrity, multi-drop bus wiring, and inter-frame silence on serial Modbus links.
4. **ASCII Mode:** Hex-encoded framing, colon delimiter, overhead comparison to RTU, and legacy HMI equipment identification.
5. **Modbus TCP:** Port 502, the MBAP header, transaction ID pipelining, and security implications of Ethernet transport without authentication.
6. **Function Codes:** Read, write, and diagnostic codes; the subset covering ninety-five percent of deployments; error response handling.
7. **Exception Codes:** Eight exception identifiers, meanings, and systematic diagnosis from exception response to root cause.
8. **Data Types:** Register-pair conventions, byte order, word order, and floating-point encoding across vendor implementations.
9. **Troubleshooting:** Addressing offset errors, byte-order mismatches, RS-485 termination faults, firewall drops, and poll rate overruns.
10. **Security:** Protocol-level limitations and compensating controls: network segmentation, unidirectional gateways, and anomaly detection.
11. **Practice Lab:** Register mapping, Exception Code 02 tracing, serial-to-TCP gateway configuration, byte-order diagnosis, and Wireshark traffic analysis.

LEARNING OUTCOMES

- Configure Modbus RTU and TCP device connections from device datasheet parameters.
- Read and interpret function code and exception code responses in a traffic capture.
- Apply byte-order and register-pair conventions to map floating-point and 32-bit values.
- Diagnose serial and TCP-layer Modbus faults using systematic isolation procedures.
- Specify network-layer compensating controls appropriate for a Modbus deployment.

REQUIRED SOFTWARE

- **ModRSim2** (free, Windows) — virtual Modbus RTU/TCP slave for register read/write exercises.
- **Modbus Poll** (trial, Windows) — master client for polling registers and inspecting exception responses.
- **Wireshark** (free, cross-platform) — packet capture for Modbus TCP traffic analysis.